

React & Web Basics – Theory Notes

What is Emmet?

- A tool that helps write HTML and CSS faster using abbreviations.
- Example: `ul>li*3` expands to a list with three list items.
- Improves speed and productivity.

Difference between Library and Framework

- Library: You control the flow and call the library when needed (e.g., React).
- Framework: Controls the flow and calls your code (e.g., Angular, Django).
- Simple idea: Library = you call it, Framework = it calls you.

What is CDN? Why do we use it?

- CDN stands for Content Delivery Network.
- It is a network of servers distributed globally.
- Provides faster loading by serving content from nearest server.
- Reduces server load and improves availability.

Why is React known as React?

- Because it reacts to changes in data.
- When state changes, UI updates automatically.
- Uses Virtual DOM to update only necessary parts.

What is crossorigin in script tag?

- Used when loading resources from another domain.
- Helps with proper error handling and security.
- Required for CORS (Cross-Origin Resource Sharing).

Difference between React and ReactDOM

- React: Core library used to create components.
- ReactDOM: Renders components to the browser DOM.

- React handles logic, ReactDOM handles rendering.

Difference between development and production builds

- Development build: Large size, includes warnings and debugging tools.
- Production build: Minified, faster, no warnings.
- Use development during coding and production for deployment.

What is async and defer?

- Both are attributes of script tag to control loading.
- async: Loads in parallel and executes immediately (order not guaranteed).
- defer: Loads in parallel but executes after HTML parsing (order maintained).